Quadratic stabilization of the canonical Foulkes–Howe map

Rewritten for readability from the OpenAI manuscript of the same title

September 25, 2026

Abstract

For every finite-dimensional complex vector space V, the canonical Foulkes–Howe map $\mathrm{Sym}^b(\mathrm{Sym}^a V) \to \mathrm{Sym}^a(\mathrm{Sym}^b V)$ is surjective whenever $a \geq 2$ and $b \geq a(a-1)$. The bound is independent of $\dim V$.

1 The result

Throughout, V is a finite-dimensional complex vector space, and products of vectors are taken in the symmetric algebra $\mathrm{Sym}\, V = \bigoplus_{j\geq 0} \mathrm{Sym}^j V$.

Averaging. For a complex vector space W and an integer $a \geq 1$, the group $\mathfrak{S}_a$ permutes the factors of $W^{\otimes a}$, and

$$\mathrm{Av}(w_1 \otimes \cdots \otimes w_a) = \frac{1}{a!} \sum_{\pi \in \mathfrak{S}_a} w_{\pi(1)} \otimes \cdots \otimes w_{\pi(a)} \tag{1}$$

defines a linear projection of $W^{\otimes a}$ onto the invariants $(W^{\otimes a})^{\mathfrak{S}_a}$. The right side is symmetric in the w_i, so $w_1 \cdots w_a \mapsto \mathrm{Av}(w_1 \otimes \cdots \otimes w_a)$ is a well-defined map $\mathrm{Sym}^a W \to (W^{\otimes a})^{\mathfrak{S}_a}$. It is an isomorphism: its inverse is the quotient map $W^{\otimes a} \to \mathrm{Sym}^a W$ restricted to invariants.

The map. Fix $a \geq 1$ and put

$$R_j = \big((\mathrm{Sym}^j V)^{\otimes a}\big)^{\mathfrak{S}_a} \quad (j \geq 0), \qquad R = \bigoplus_{j\geq 0} R_j.$$

The algebra $(\mathrm{Sym}\, V)^{\otimes a}$, with factorwise multiplication, is commutative, and $\mathfrak{S}_a$ acts on it by algebra automorphisms; hence R is a graded commutative subalgebra. Averaging identifies $R_1 = \mathrm{Sym}^a V$ and $R_b = \mathrm{Sym}^a(\mathrm{Sym}^b V)$. For $b \geq 0$, multiplying b elements of R_1 gives a linear map $\mathrm{Sym}^b R_1 \to R_b$. Under these identifications it is the *canonical Foulkes–Howe map*

$$\mu_{a,b,V} : \mathrm{Sym}^b(\mathrm{Sym}^a V) \longrightarrow \mathrm{Sym}^a(\mathrm{Sym}^b V). \tag{2}$$

Its image is A_b, the degree-b part of the subalgebra $A \subseteq R$ generated by R_1. (In positive dimension, Raicu–Sam–Weyman [11, Section 2] formulate this map as $\mathrm{Sym}^b(D^a V) \to D^a(\mathrm{Sym}^b V)$, with divided powers realized as invariant tensors; averaging transports it to (2).)

Theorem 1. *For every integer $a \geq 2$, every integer $b \geq a(a-1)$, and every finite-dimensional complex vector space V, the map $\mu_{a,b,V}$ in (2) is surjective. Equivalently, $A_b = R_b$.*

For $a = 1$ the map is the identity in every degree. Corollary 8 deduces a $\mathrm{GL}(V)$-equivariant embedding $\mathrm{Sym}^a(\mathrm{Sym}^b V) \hookrightarrow \mathrm{Sym}^b(\mathrm{Sym}^a V)$ in the same range.

Proof at a glance. The idea is to dualize surjectivity into a vanishing statement for symmetric multilinear forms, and then to deduce that a polynomial is zero from the vanishing of a chain of its directional derivatives.

1. *Criterion* (Section 2). A_b is spanned by the tensors $P^{\otimes a}$ with $P = u_1 \cdots u_b$ a product of b vectors (Lemmas 2, 3). Dually (Proposition 4), $\mu_{a,b,V}$ is surjective if and only if every symmetric a-linear form T on $\operatorname{Sym}^b V$ with $T(P, \dots, P) = 0$ for all such P is zero.
2. *Injectivity* (Section 3). The derivative $D_{Y,U} = \sum_k Y_k \, \partial/\partial U_k$ is injective on polynomials whose U-degree exceeds their Y-degree (Lemma 5). So a polynomial is zero if a chain of such derivatives of it is zero and the degree inequality holds before each derivative (Corollary 6).
3. *Vanishing* (Proposition 7, Section 4). If $b \geq a(a-1)$, every T as in item 1 is zero. The proof is by induction on a; write $r = a - 1$ and $m = b - r \geq r^2$.
   - Step 1: test T on $P = (x^r + ct^r)Q$, which is a product of b vectors because binary forms split over $\mathbb{C}$. The coefficient of c gives $T(x^rQ, \dots, x^rQ, t^rQ) = 0$.
   - Step 2: m derivatives turn t^b into a multiple of t^rQ, so Corollary 6 gives $T(x^rQ, \dots, x^rQ, t^b) = 0$.
   - Step 3: the induction hypothesis, applied to the first r slots, replaces the common Q by arbitrary $Q_1, \dots, Q_r \in \operatorname{Sym}^m V$.
   - Step 4: r^2 derivatives turn $T(v_1^b, \dots, v_r^b, t^b)$ into a multiple of $T(x^r v_1^m, \dots, x^r v_r^m, t^b) = 0$. They all raise the degree of the same block x, up to $r^2 - 1$, while the source degrees stay $\geq m + 1$. This accumulation produces the quadratic bound.

Dependencies: Lemma 2 → Lemma 3 → Proposition 4; Lemma 5 → Corollary 6 → Steps 2 and 4; Steps 1 → 2 → 3 → 4; Propositions 4 and 7 → Theorem 1 → Corollary 8.

Context. The affine Chow variety of decomposable a-forms consists of products of a linear forms in $\operatorname{Sym}^a(V^*)$; its coordinate ring identifies with A, whose normalization is R, with b the coordinate degree [11, Section 2, Theorem 2.3]; see also [8, Section 7.6, Propositions 7.11–7.12]. Brion proved eventual surjectivity [2] and then an effective bound involving both a and $\dim V$ [3, Theorem 3.3]. Raicu, Sam, and Weyman bound the regularity of the Segre ring and modules of covariants, including the normalization [11, Theorem 3.9]; this does not compute the top degree of the quotient R/A, which controls stabilization. Theorem 1 gives the bound $a(a-1)$ in every dimension and answers positively the request for a polynomial bound in [8, Problem 7.19] (there $d = b$, $n = a$, $w = \dim V$; the bound is $d \geq n(n-1)$, including $w = n$). The bound is not asserted to be sharp. The question differs from the Foulkes conjecture [5], which asks, for $a \leq b$, for some $\mathrm{GL}(V)$-equivariant injection $\operatorname{Sym}^a(\operatorname{Sym}^b V) \hookrightarrow \operatorname{Sym}^b(\operatorname{Sym}^a V)$ [11, Conjecture 2.10]. Surjectivity of the canonical map implies such an injection (Corollary 8), but an injection need not come from the canonical map: at $a = b$ the two representations coincide, yet the canonical map has nonzero kernel at $(a, b) = (5, 5)$ and $(6, 6)$ in suitable dimensions [4, arXiv version, Theorem 6(b),(c) and Remark 7], both outside our range. Successive weight shifts also occur in Ikenmeyer's proof [7, Section 5] of McKay's propagation theorem [10]: for the opposite canonical maps $\operatorname{Sym}^a(\operatorname{Sym}^b V) \to \operatorname{Sym}^b(\operatorname{Sym}^a V)$ with $\dim V \geq b$, injectivity at (a, B), $B \geq a$, propagates to all (a, b) with $b \geq B$ [7, Theorem 1.3 and Section 3].

2 The canonical image and diagonal vanishing

Fix integers $a, b \geq 1$ and put $E = \operatorname{Sym}^b V$. A *product of b vectors* is an element $P = u_1 \cdots u_b \in E$ with $u_i \in V$; the vectors may repeat or be zero.

Lemma 2. *For every integer $k \geq 1$, the pure powers u^k $(u \in V)$ span $\operatorname{Sym}^k V$.*

This is used in Lemma 3 ($k = a$) and at the end of the proof of Proposition 7 ($k = b$).

Proof. We claim the polarization identity

$$k!\, u_1 \cdots u_k = \sum_{J \subseteq \{1,\dots,k\}} (-1)^{k-|J|} \Big(\sum_{j \in J} u_j \Big)^k .$$

Treat the u_i as formal commuting variables and expand. A degree-k monomial with support $S \subseteq \{1,\dots,k\}$ occurs in $(\sum_{j\in J} u_j)^k$ exactly when $J \supseteq S$, with a multinomial coefficient γ independent of J. Its total coefficient is $\gamma \sum_{J \supseteq S} (-1)^{k-|J|} = \gamma(1-1)^{k-|S|}$. This is 0 unless $S = \{1,\dots,k\}$, in which case the monomial is $u_1 \cdots u_k$ and $\gamma = k!$. Substituting vectors gives the identity in $\operatorname{Sym} V$. Products span $\operatorname{Sym}^k V$ and $k! \neq 0$, so pure powers span. □

Lemma 3. *$A_b = \operatorname{span}_{\mathbb{C}}\{P^{\otimes a} : P$ a product of b vectors$\}$.*

This is used only in Proposition 4.

Proof. (i) Averaging sends $u^a \in \operatorname{Sym}^a V$ to $z(u) := u^{\otimes a}$. Averaging maps $\operatorname{Sym}^a V$ onto R_1, so by Lemma 2 with $k = a$ the elements $z(u)$ span R_1. (ii) A_b is spanned by products of b elements of R_1. Multiplication is multilinear, so A_b is spanned by the products $z(u_1) \cdots z(u_b)$. (iii) Factorwise multiplication gives $z(u_1) \cdots z(u_b) = (u_1 \cdots u_b)^{\otimes a}$. □

Proposition 4 (Diagonal criterion)**.** *The map $\mu_{a,b,V}$ is surjective if and only if every symmetric a-linear form T on $E = \operatorname{Sym}^b V$ satisfying $T(P,\dots,P) = 0$ for all products P of b vectors is zero.*

The proof of Theorem 1 uses the "if" direction.

Proof. For $\lambda \in R_b^*$ define the symmetric a-linear form $T_\lambda(w_1,\dots,w_a) = \lambda\big(\mathrm{Av}(w_1 \otimes \cdots \otimes w_a)\big)$ on E.

(i) *$\lambda \mapsto T_\lambda$ is a bijection from R_b^* onto the symmetric a-linear forms on E.* It is injective because Av maps $E^{\otimes a}$ onto R_b, so averages of simple tensors span R_b. It is surjective: given a symmetric T, let $\widetilde{T} : E^{\otimes a} \to \mathbb{C}$ be its linearization and $\lambda = \widetilde{T}|_{R_b}$. Then $T_\lambda(w_1,\dots,w_a) = \frac{1}{a!}\sum_\pi T(w_{\pi(1)},\dots,w_{\pi(a)}) = T(w_1,\dots,w_a)$ by symmetry.

(ii) For a product P, the tensor $P^{\otimes a}$ is invariant, so $T_\lambda(P,\dots,P) = \lambda(P^{\otimes a})$. By Lemma 3, λ vanishes on A_b if and only if $T_\lambda(P,\dots,P) = 0$ for all products P.

(iii) R_b is finite-dimensional, so $A_b = R_b$ if and only if the only $\lambda \in R_b^*$ vanishing on A_b is 0. By (i) and (ii) this is the stated condition on forms, and $\operatorname{im} \mu_{a,b,V} = A_b$. □

3 Differential injectivity

Conventions. Assume $n = \dim V > 0$ and fix a basis $e_1,\dots,e_n$. A *vector variable* is a block $U = (U_1,\dots,U_n)$ of independent scalar variables, standing for $\sum_k U_k e_k$. An expression built from vector variables by products in $\operatorname{Sym} V$ and a multilinear form expands, by multilinearity, to a polynomial in the coordinates, homogeneous in each block. A polynomial over $\mathbb{C}$ that vanishes at every point is zero (induction on the number of variables, from the univariate root bound). Hence such a polynomial is zero once the expression vanishes for all vector values.

For two blocks U, Y define

$$D_{Y,U} = \sum_{k=1}^{n} Y_k \frac{\partial}{\partial U_k}.$$

It lowers the U-degree by one, raises the Y-degree by one, and leaves the degree in every other block unchanged. By the chain rule, $D_{Y,U} h = \frac{d}{ds} h(U + sY)\big|_{s=0}$, with the other blocks held fixed. Let L

be a linear form on $\mathrm{Sym}^{k+j} V$ and $g \in \mathrm{Sym}^j V$, neither involving U (they may depend polynomially on other blocks, including Y). Since $(U+sY)^k g = U^k g + sk\, U^{k-1} Y g + O(s^2)$ in $\mathrm{Sym}\, V$,

$$D_{Y,U}\, L(U^k g) = k\, L(U^{k-1} Y\, g) \qquad (k \geq 1). \tag{3}$$

Lemma 5. *Let p, q be nonnegative integers. On polynomials homogeneous of degree p in U and degree q in Y, the operator $D_{Y,U}$ is injective if $p > q$. The polynomials may depend on any finite collection of additional variables independent of U and Y.*

This is the positive-weight injectivity in $\mathfrak{sl}_2$ weight strings; compare [6, Section 11.1, Claim 11.4 and equation (11.5)]. Ikenmeyer's argument [7, Section 5, Claim 5.1] decomposes tensors according to untouched positions before applying a two-coordinate weight shift. The proof below is a direct inner-product argument with all additional variables present. The condition $p > q$ cannot be dropped: for $p = q = 1$, $h = U_1 Y_2 - U_2 Y_1$ satisfies $D_{Y,U} h = 0$. (Renaming: the source calls the operator $D_{U,Y}$ below E; we write D', since E denotes $\mathrm{Sym}^b V$.)

Proof. Let X list all coordinates, including the additional variables. Use the Fischer (apolar) Hermitian inner product [1, Section 2, equation (8) and Proposition 3]: distinct monomials are orthogonal and $\|X^\alpha\|^2 = \alpha! := \prod_k \alpha_k!$. It is positive definite.

Adjoints. For each coordinate X_j, ∂_{X_j} is adjoint to multiplication by X_j: on monomials, $\langle \partial_{X_j} X^\alpha, X^\beta \rangle$ and $\langle X^\alpha, X_j X^\beta \rangle$ both equal $\alpha!$ if $\alpha = \beta + e_j$ and 0 otherwise (using $\alpha_j \cdot (\alpha - e_j)! = \alpha!$). Put $D = D_{Y,U}$ and $D' = D_{U,Y}$. Then $(Y_k \partial_{U_k})^* = (\partial_{U_k})^* (Y_k)^* = U_k \partial_{Y_k}$, so $D^* = D'$ and $D'^* = D$.

Commutator. By the Leibniz rule,

$$U_j \partial_{Y_j}\, Y_k \partial_{U_k} = \delta_{jk} U_j \partial_{U_k} + U_j Y_k \partial_{Y_j} \partial_{U_k}, \qquad Y_k \partial_{U_k}\, U_j \partial_{Y_j} = \delta_{jk} Y_k \partial_{Y_j} + Y_k U_j \partial_{U_k} \partial_{Y_j}.$$

Subtracting and summing over j, k,

$$D'D - DD' = N_U - N_Y, \qquad N_U = \sum_k U_k \partial_{U_k}, \quad N_Y = \sum_k Y_k \partial_{Y_k}.$$

On a monomial, N_U multiplies by the U-degree and N_Y by the Y-degree.

Conclusion. For h of bidegree (p, q) in (U, Y),

$$\|Dh\|^2 - \|D'h\|^2 = \langle D'Dh, h \rangle - \langle DD'h, h \rangle = \langle (N_U - N_Y)h, h \rangle = (p - q)\|h\|^2. \tag{4}$$

If $Dh = 0$ and $p > q$, then $(p - q)\|h\|^2 = -\|D'h\|^2 \leq 0$, so $h = 0$. □

Corollary 6. *Let G_0 be a polynomial homogeneous in each block, and let $G_i = D_{Y^{(i)},U^{(i)}} G_{i-1}$ for $1 \leq i \leq N$. Suppose that $G_N = 0$ and that, for every i, the degree of G_{i-1} in $U^{(i)}$ exceeds its degree in $Y^{(i)}$. Then $G_0 = 0$.*

Proof. Each G_{i-1} is homogeneous in each block. Lemma 5 applied to G_{N-1} gives $G_{N-1} = 0$; repeat downward to G_0. □

Corollary 6 is used in Steps 2 and 4 below.

4 Vanishing on products

Proposition 7. *Let V be a finite-dimensional complex vector space, and let $a \geq 1$ and $b \geq 1$ be integers satisfying $b \geq a(a-1)$. If a symmetric a-linear form T on $\operatorname{Sym}^b V$ satisfies*

$$T(P, \dots, P) = 0 \qquad \text{for every product } P \text{ of } b \text{ vectors}, \tag{5}$$

then $T = 0$.

Plan. We induct on a. In the inductive step, Step 1 produces a vanishing with a common factor Q in every slot, Step 2 removes Q from the last slot, Step 3 applies the induction hypothesis to the other $r = a - 1$ slots, and Step 4 removes the common power x^r from those slots.

Proof. If $V = 0$ then $\operatorname{Sym}^b V = 0$ and $T = 0$. So let $\dim V > 0$ and use the conventions of Section 3. We induct on a; the statement for a given a covers every admissible b and every V.

Base case $a = 1$. T is a linear form vanishing on all products of b vectors, and these span $\operatorname{Sym}^b V$.

Inductive step. Let $a \geq 2$, assume the statement for $a - 1$, and put

$$r = a - 1 \geq 1, \qquad m = b - r. \tag{6}$$

Since $b \geq a(a-1) = r(r+1)$,

$$m \geq r^2 \geq 1. \tag{7}$$

Step 1: a split binary form. *Claim: for all $x, t \in V$ and every product $Q = y_1 \cdots y_m$ of m vectors,*

$$T(\underbrace{x^r Q, \dots, x^r Q}_{r \text{ slots}}, t^r Q) = 0. \tag{8}$$

(a) *For every $c \in \mathbb{C}$, $P_c = (x^r + ct^r)Q$ is a product of b vectors.* Choose $\alpha \in \mathbb{C}$ with $\alpha^r = -c$ and let $\zeta = e^{2\pi i/r}$. From $\prod_{j=0}^{r-1}(X - \beta\zeta^j) = X^r - \beta^r$ with $\beta = \alpha Y$,

$$X^r + cY^r = \prod_{j=0}^{r-1}(X - \alpha\zeta^j Y) \quad \text{in } \mathbb{C}[X, Y].$$

The algebra homomorphism $\mathbb{C}[X, Y] \to \operatorname{Sym} V$, $X \mapsto x$, $Y \mapsto t$ (which needs no independence of x, t) writes $x^r + ct^r$ as a product of r vectors, and $r + m = b$.

(b) By (a) and (5), $T(P_c, \dots, P_c) = 0$ for all c. Since $P_c = x^r Q + c\, t^r Q$, multilinearity and symmetry give

$$T(P_c, \dots, P_c) = \sum_{k=0}^{a} \binom{a}{k} c^k\, T(\underbrace{x^r Q, \dots, x^r Q}_{a-k}, \underbrace{t^r Q, \dots, t^r Q}_{k}).$$

A polynomial in c vanishing on $\mathbb{C}$ has zero coefficients. The coefficient of c is $a\, T(x^r Q, \dots, x^r Q, t^r Q)$ with $a - 1 = r$ slots $x^r Q$, and $a \neq 0$.

Step 2: freeing the last slot. *Claim: for all $x, t \in V$ and every product Q of m vectors, $T(x^r Q, \dots, x^r Q, t^b) = 0$ (r slots $x^r Q$).*

Fix $x \in V$. Let $t, y_1, \dots, y_m$ be vector variables, $Q = y_1 \cdots y_m$, and

$$G_0 = H_x = T(\underbrace{x^r Q, \dots, x^r Q}_{r \text{ slots}}, t^b), \qquad G_i = D_{y_i,t} G_{i-1} \quad (1 \le i \le m).$$

(a) *Formula.* For $0 \le i \le m$,

$$G_i = \frac{b!}{(b-i)!}\, T(x^r Q, \dots, x^r Q,\ t^{b-i} y_1 \cdots y_i). \tag{9}$$

This follows by induction on i from (3) with $U = t$, $Y = y_i$, $k = b - i + 1$, $g = y_1 \cdots y_{i-1}$ and $L = T(x^r Q, \dots, x^r Q, \cdot\,)$; the first r slots do not involve t.

(b) $G_m = 0$. Since $b - m = r$, G_m is a multiple of $T(x^r Q, \dots, x^r Q, t^r Q)$. This vanishes for all vector values by (8), hence as a polynomial.

(c) *Degrees.* By (9), G_{i-1} has t-degree $b - i + 1$ and y_i-degree r: y_i occurs once in each of the r copies of Q and not yet in the last slot. The difference is $(b - i + 1) - r = m - i + 1 \ge 1$.

(d) Corollary 6 gives $H_x = G_0 = 0$, which is the claim.

Step 3: induction on the remaining slots. *Claim: for all $x, t \in V$,*

$$T(x^r Q_1, \dots, x^r Q_r, t^b) = 0 \qquad (Q_1, \dots, Q_r \in \operatorname{Sym}^m V). \tag{10}$$

Fix x, t. Multiplication by x^r is linear $\operatorname{Sym}^m V \to \operatorname{Sym}^b V$, so $S_{x,t}(Q_1, \dots, Q_r) := T(x^r Q_1, \dots, x^r Q_r, t^b)$ is a symmetric r-linear form on $\operatorname{Sym}^m V$. By Step 2, $S_{x,t}(Q, \dots, Q) = 0$ for every product Q of m vectors. The induction hypothesis applies with (r, m) in place of (a, b), because $r \ge 1$, $m \ge 1$ and $m \ge r^2 \ge r(r-1)$ by (7). Hence $S_{x,t} = 0$.

Step 4: removing the common power. *Claim: $T(v_1^b, \dots, v_r^b, t^b) = 0$ for all $v_1, \dots, v_r, t \in V$.*

Fix $t \in V$. Let $v_1, \dots, v_r$ and x be vector variables (so x is now a variable block) and

$$f = T(v_1^b, \dots, v_r^b, t^b),$$

of degree b in each v_i and degree 0 in x. Apply r^2 operators in this order: D_{x,v_1} r times, then D_{x,v_2} r times, and so on up to D_{x,v_r}.

(a) *Result.* Apply (3) r times in slot i, with $U = v_i$, $Y = x$ and g a power of x; the other slots do not involve v_i. This replaces v_i^b by $b(b-1)\cdots(b-r+1)\, x^r v_i^{b-r} = \frac{b!}{m!} x^r v_i^m$. Hence

$$D_{x,v_r}^r \cdots D_{x,v_1}^r f = \left(\frac{b!}{m!}\right)^r T(x^r v_1^m, \dots, x^r v_r^m, t^b) = 0. \tag{11}$$

The last equality holds for all vector values by (10) with $Q_i = v_i^m$, hence as a polynomial identity.

(b) *Degrees.* Before the j-th of the r^2 operators, the x-degree is $j - 1 \le r^2 - 1$. The source block v_i of that operator has been differentiated at most $r - 1$ times, so its degree is at least $b - (r-1) = m + 1 \ge r^2 + 1$ by (7). So the source degree exceeds the x-degree.

(c) Corollary 6 gives $f = 0$.

Conclusion. By Step 4, T vanishes on every a-tuple of pure b-th powers. These span $\operatorname{Sym}^b V$ (Lemma 2, $k = b$), so $T = 0$ by multilinearity. This completes the induction. □

5 Proof of the theorem; representation comparison

Proof of Theorem 1. Let $a \geq 2$ and $b \geq a(a-1)$; then $b \geq 1$. Proposition 7 shows that every symmetric a-linear form on $\mathrm{Sym}^b V$ whose diagonal vanishes on products of b vectors is zero. Proposition 4 then gives surjectivity of $\mu_{a,b,V}$. □

Corollary 8. *For every finite-dimensional complex vector space V, every integer $a \geq 2$, and every integer $b \geq a(a-1)$, there exists a $\mathrm{GL}(V)$-equivariant embedding*

$$\mathrm{Sym}^a(\mathrm{Sym}^b V) \hookrightarrow \mathrm{Sym}^b(\mathrm{Sym}^a V).$$

In particular, this gives the sixth symmetric-power comparison for $b \geq 30$.

Proof. Averaging and multiplication commute with the action of $\mathrm{GL}(V)$, so the surjection $\mu_{a,b,V}$ of Theorem 1 is equivariant. Finite-dimensional polynomial representations over $\mathbb{C}$ are completely reducible [9, Chapter I, Appendix A, Sections 7–8]. Choose an invariant complement M to $\ker \mu_{a,b,V}$. Then $\mu_{a,b,V}|_M$ is an equivariant isomorphism onto $\mathrm{Sym}^a(\mathrm{Sym}^b V)$, and its inverse followed by the inclusion $M \subseteq \mathrm{Sym}^b(\mathrm{Sym}^a V)$ is the embedding. This proves existence only; the embedding is not identified with a prescribed canonical map. □

Remark 9. Lemma 5 holds for any number of coordinates and additional variables, and the degree comparisons involve only a and b. With the finite-dimensional annihilator argument, this gives the same bound in every finite dimension, without a stable-dimension reduction. The proof uses positivity of the Fischer inner product, splitting of binary forms over $\mathbb{C}$, and the nonzero characteristic-zero scalars $(a,\ k!,\ b!/(b-i)!,\ b!/m!)$ in the coefficient and differentiation identities.

6 Verification checklist

1. **Duality** (Proposition 4, (i)–(ii)): the restricted linearization λ satisfies $T_\lambda = T$, and $T_\lambda(P,\dots,P) = \lambda(P^{\otimes a})$ with $P^{\otimes a}$ spanning A_b (Lemma 3).
2. **Injectivity lemma** (Lemma 5): the adjointness $D^* = D'$, the commutator $D'D - DD' = N_U - N_Y$, and the sign in (4).
3. **Step 1(a)**: $(x^r + ct^r)Q$ is a product of exactly $r + m = b$ vectors for every c and all x, t, including dependent ones. **Step 1(b)**: the coefficient of c is $a\,T(x^rQ,\dots,x^rQ,t^rQ)$.
4. **Step 2(c)**: before the i-th derivative, the y_i-degree is exactly r and the t-degree is $b-i+1$, so the difference is $m-i+1 \geq 1$.
5. **Step 3**: the induction hypothesis is applied to (r,m) and needs $m \geq 1$ and $m \geq r(r-1)$; both follow from (7).
6. **Step 4(b)**: all r^2 derivatives raise the degree of the same block x (at most r^2-1 beforehand), while each source degree is at least $m+1 \geq r^2+1$. Check also the constant $(b!/m!)^r \neq 0$ in (11).
7. **Polynomial versus pointwise vanishing**: Steps 2(b) and 4(a) pass from vanishing at all vector values to a polynomial identity (Conventions, Section 3), which Corollary 6 requires.

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